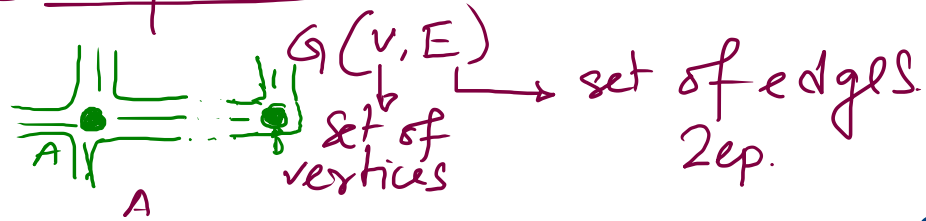
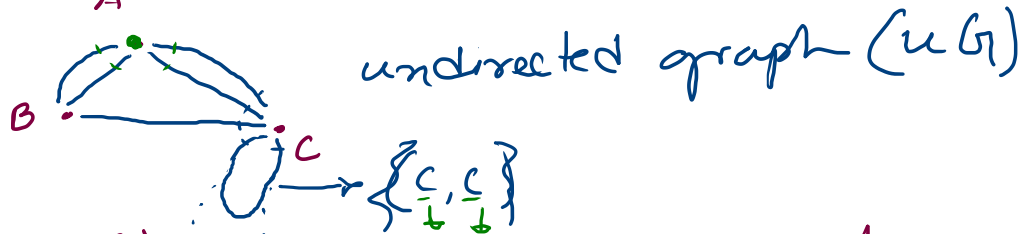


10.1 Graphs and features.

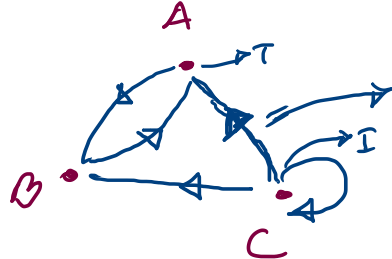
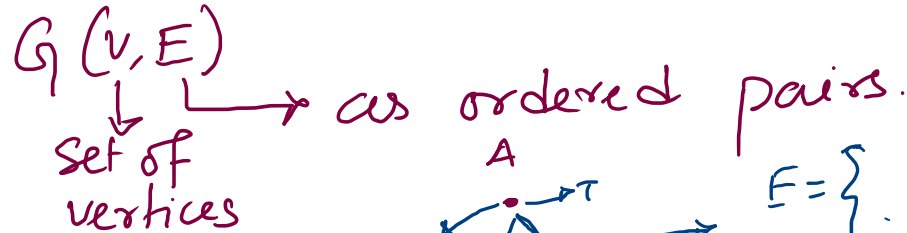
⑩ Defⁿ 1:



V $\begin{cases} \text{finite} \rightarrow \text{finite graphs.} \\ \text{infinite} \rightarrow \text{infinite graphs.} \end{cases}$

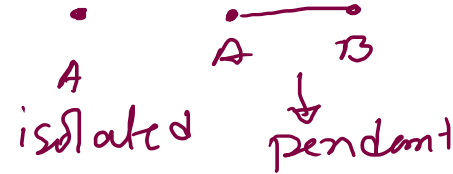


⑩ Defⁿ 2: Directed graphs/digraph (dG)



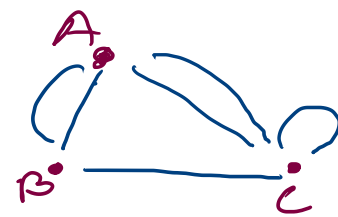
$$E = \{ \dots (C, A) \dots \}$$

vertex
 $\rightarrow$ degree 1 $\rightarrow$ pendant
 $\rightarrow$ degree 0 $\rightarrow$ isolated



* Th 1: Hand-shaking Theorem.

$G(V, E) \rightarrow m$ edges. then $\underline{2m} = \sum_{v \in V} \deg(v)$



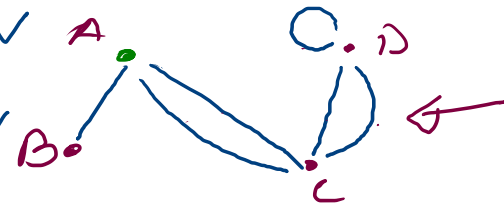
$m=6$
 $\deg(A)=4$
 $\deg(B)=3$
 $\deg(C)=5$

 $\sum \deg(v) = 12$
 $2m = 12$

* Th 2: An ug has even no. of vertices of odd degree

degree
 $2m = \sum_{v \in V} \deg(v)$

$\underbrace{V}_{\text{even degree}} \rightarrow v_1 \in V$
 $\underbrace{V}_{\text{odd degree}} \rightarrow v_2 \in V$



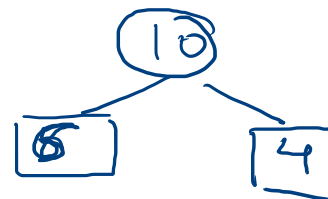
$= \sum_{v_1 \in V_1} \deg(v_1) + \sum_{v_2 \in V_2} \deg(v_2)$

① $2m \rightarrow \text{even.}$
 $\sum_{v_1 \in V_1} \deg(v_1) \rightarrow \text{even.}$

eg: $2m=12$

even_deg $\rightarrow C, D$

odd_deg $\rightarrow A, B$



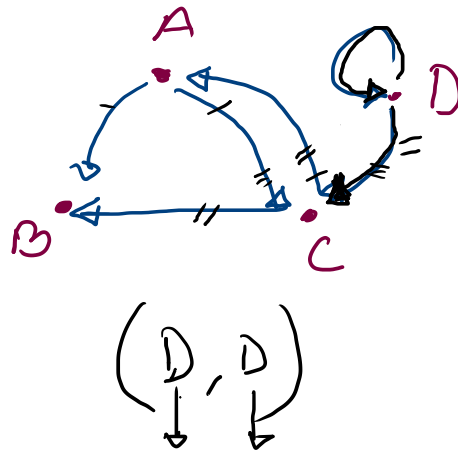
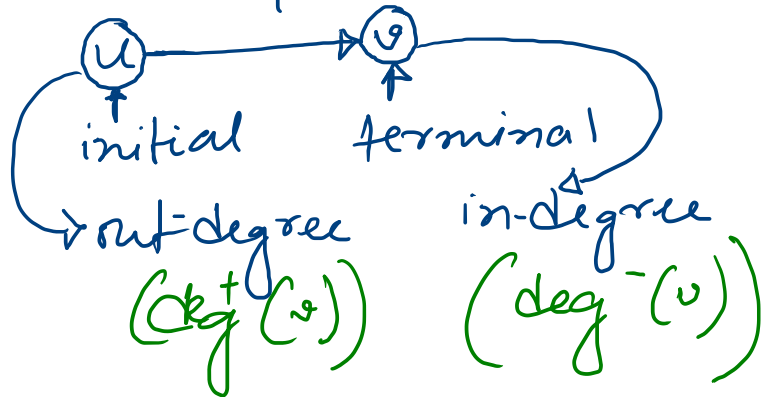
$0 + 5 + 3 = 8$
 $4 + 3 = 7$
 $4 + 3 + 1 = 8$
 2

$\sum_{v_1 \in V_1} \deg(v_1) + 7 = 2m$

④ Defn 5

$$dG \rightarrow G(V, E)$$

ordered pair (edge)



v	$\deg^-(v)$	$\deg^+(v)$
A	1	2
B	2	0
C	2	2
D	1	2
Σ	<u>6</u>	<u>6</u>

$$|E| = m = 6$$

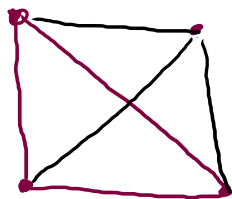
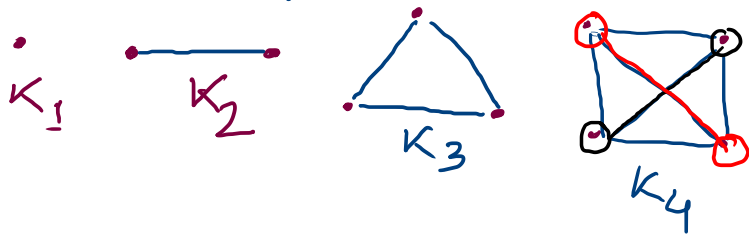
④ Th-3 in a dig

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

* Types of graphs.

① Complete graphs. (K_n)

A graph with exactly one edge b^{tw} each pair of distinct vertices.

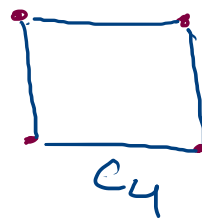
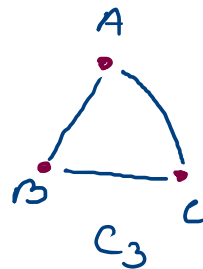


$$K_4 \equiv W_3$$

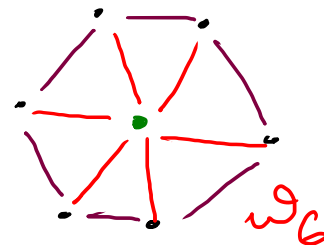
② Cycles ($n \geq 3$)

$$G(V, E) \rightarrow C_n$$

$$\{ \{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\} \}$$



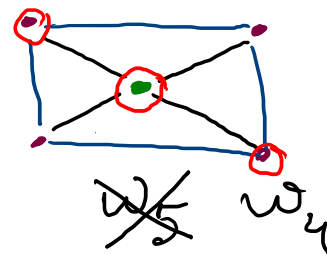
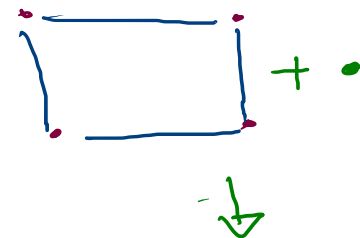
$$W_6 = C_6 + 1$$



③ Wheels

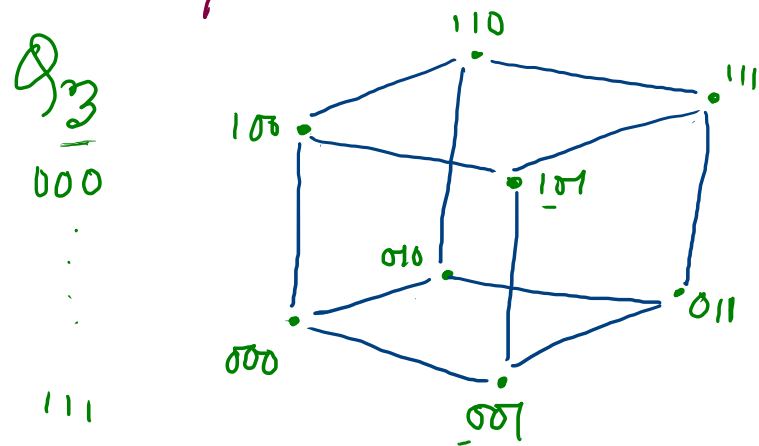
$$W_n = C_n + 1$$

$$W_4 = C_4 + 1$$



④ n-cubes $\rightarrow$ n-dimensional hypercubes.
 $\downarrow$
 Q_n

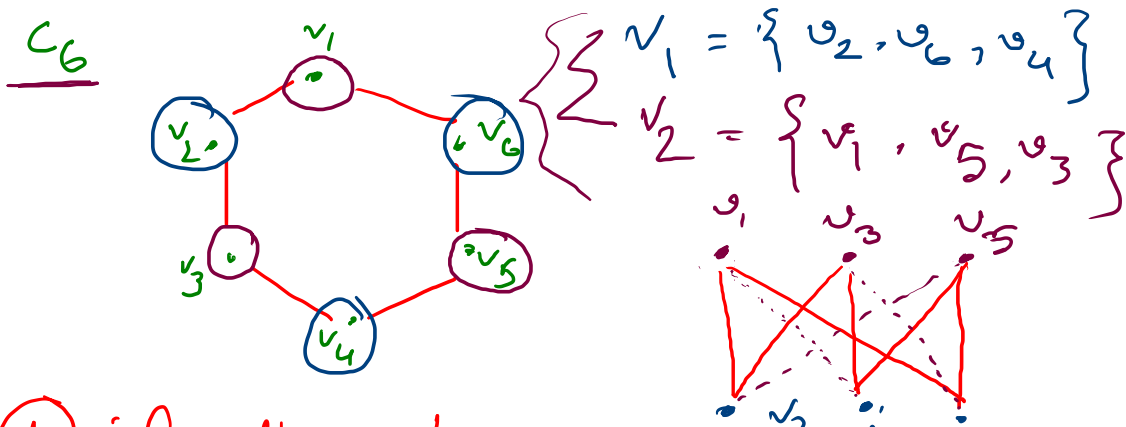
- Vertices $\rightarrow$ represented using 2^n bit strings of length n.
- 2 vertices are adjacent iff their bit strings differ by exactly one bit.



⑤ Bipartite graphs.

$$G(V, E) \rightarrow V \begin{matrix} \swarrow \\ v_1 \\ \searrow \\ v_2 \end{matrix} \rightarrow \underbrace{v_1 \cap v_2 = \emptyset}_{\text{are disjoint}}$$

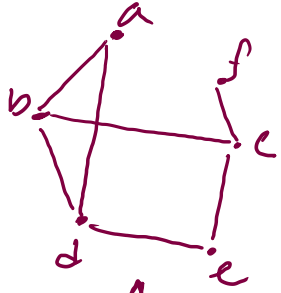
If V can be divided into 2 disjoint sets V_1, V_2 such that there are no edges among the elements of V_1 or V_2 those graphs are bipartite graphs.



* if all vertices in both sets are interconnected then $\rightarrow$ complete bipartite

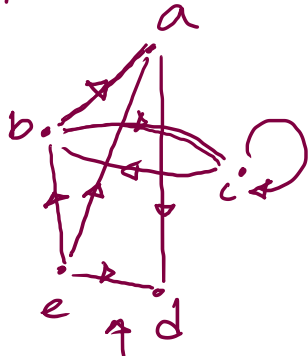
10.3 Graph representation

① Adjacency list



UG

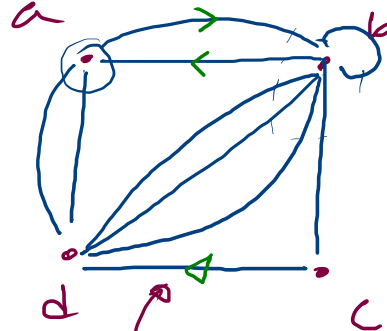
V	V _A
a	b, d
b	a, c, d
c	b, e, f
d	a, b, e
e	c, d
f	c



DG

V _i	V _j
a	d
b	a, c
c	c, b
d	-
e	b, a, d

Adjacent Matrix

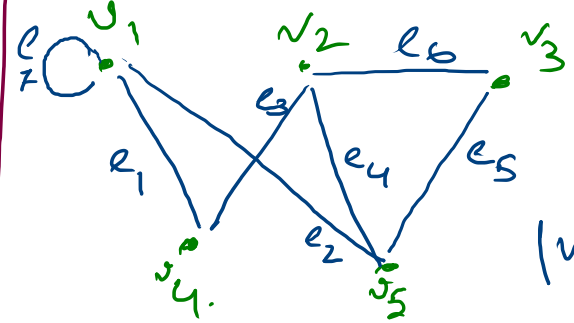


$|V| \times |V|$

	a	b	c	d
a	0	1	2	0
b	2	1	1	3
c	0	1	0	1
d	2	3	0	0

$M_{0,0} = 1 \rightarrow \text{self loop}$

Incidence Matrix



$|V| \times |E|$

	e ₁	e ₂	e ₃	e ₄	e ₅	e ₆	e ₇
v ₁	1	1	0	0	0	0	1
v ₂	0	0	1	1	0	1	0
v ₃	0	0	0	0	1	1	0
v ₄	1	0	1	1	0	0	0
v ₅	0	1	0	1	1	0	0

$M_{e_7, v_1} = 1, \text{ rest } = 0$